



CS1002 –
Programming
Fundamentals
Assignment # 05

Instructor	Ch. Usman Ghous
Session	Fall 2023
Section	BCS 1B

General Guidelines

1. Peer plagiarism and late submissions are strictly not allowed. In case, zero marks will be awarded for whole assignment.
2. Total Marks: 120

Submission Guidelines

1. You will upload the assignment on CLASSROOM in the given timeline.
2. Don't email your solution to the instructor or TA for submission. Submit your assignment in the given deadline, said LMS.
3. No submission will be accepted later than said deadline.

Deadline: December 18, 2023, 9:00

Q.No.1:

Roman numerals are represented by seven different symbols: I, V, X, L, C, D and M.

Symbol	Value
I	1
V	5
X	10
L	50
C	100
D	500
M	1000

For example, 2 is written as II in Roman numeral, just two one's added together. 12 is written as XII, which is simply X + II. The number 27 is written as XXVII, which is XX + V + II.

Roman numerals are usually written largest to smallest from left to right. However, the numeral for four is not IIII. Instead, the number four is written as IV. Because the one is before the five we subtract it making four. The same principle applies to the number nine, which is written as IX. There are six instances where subtraction is used:

- I can be placed before V (5) and X (10) to make 4 and 9.
- X can be placed before L (50) and C (100) to make 40 and 90.
- C can be placed before D (500) and M (1000) to make 400 and 900.

Given an integer, convert it to a roman numeral.

Example 1:

Input: num = 3
Output: "III"
Explanation: 3 is represented as 3 ones.

Example 2:

Input: num = 58
Output: "LVIII"
Explanation: L = 50, V = 5, III = 3.

Example 3:

Input: num = 1994
Output: "MCMXCIV"
Explanation: M = 1000, CM = 900, XC = 90 and IV = 4.

Q.No.2:

Given an integer array `nums` sorted in **non-decreasing order**, remove the duplicates in-place such that each unique element appears only **once**. The **relative order** of the elements should be kept the **same**. Then return the number of unique elements in `nums`.

Q.No.3:

Given an integer n , return *the number of prime numbers that are strictly less than n* . present in array, also print the prime number

Example 1:

Input: $n = 10$

Output: 4

Q.No.4:

Given an integer array `nums`, move all 0's to the end of it while maintaining the relative order of the non-zero elements.

Note that you must do this in-place without making a copy of the array.

Example 1:

Input: `nums = [0,1,0,3,12]`

Output: `[1,3,12,0,0]`

Q.No.5:

A **perfect number** is a **positive integer** that is equal to the sum of its **positive divisors**, excluding the number itself. A **divisor** of an integer x is an integer that can divide x evenly.

Given an integer n , return `true` *if n is a perfect number*, otherwise return `false`.

Example 1:

Input: `num = 28`

Output: `true`

Explanation: `28 = 1 + 2 + 4 + 7 + 14`

1, 2, 4, 7, and 14 are all divisors of 28.

Q.No.6:

Write a program that uses while loops to perform the following steps:

- Prompt the user to input two integers: `firstNum` and `secondNum` (`firstNum` must be less than `secondNum`).
- Output all odd numbers between `firstNum` and `secondNum`.
- Output the sum of all even numbers between `firstNum` and `secondNum`.
- Output the numbers and their squares between 1 and 10.
- Output the sum of the square of the odd numbers between `firstNum` and

secondNum.

f) Output all uppercase letters

Q.No.7:

Suppose you give a dinner party for six guests, but your table seats only four. In how many ways can four of the six guests arrange themselves at the table? Any of the six guests can sit in the first chair. Any of the remaining five can sit in the second chair. Any of the remaining four can sit in the third chair, and any of the remaining three can sit in the fourth chair. (The last two will have to stand.) So the number of possible arrangements of six guests in four chairs is $6*5*4*3$, which is 360. Write a program that calculates the number of possible arrangements for any number of guests and any number of chairs. (Assume there will never be fewer guests than chairs.) Don't let this get too complicated. A simple for loop/s should do it.

Q.No.8:

Write a program to print the following pyramid using loops.

Note: User enters number of rows only.

```
      1
    2 3 2
  3 4 5 4 3
4 5 6 7 6 5 4
5 6 7 8 9 8 7 6 5
```

Q.No.9:

Write a C++ Program to Put Even & Odd Elements of an Array in 2 Separate Arrays. The program first finds the odd and even elements of the array. Then the odd elements of an array are stored in one array and even elements of an array is stored in another array.

Problem Solution

1. Create three integer arrays. First to store all the elements. Second to store even elements of first array. Third to store odd elements of first array.

Assumption: take a fixed size array for original elements of even size e.g. 10,

Take two more arrays of size half than the size of original array.

2. Using for loop, scan each and every element of the first array, check whether they are even or odd by taking modulo of 2 on those numbers.

3. Even numbers will get stored(copied) in second array, while odd numbers will get stored in third array.
4. Print both arrays.

Q.No.10:

Write a C++ Program to Sort N Numbers in Ascending Order using Bubble Sort.

Bubble sort is a simple sorting algorithm that works by repeatedly stepping through the list to be sorted, comparing each pair of adjacent items and swapping them if they are in the wrong order. Here we need to sort a number in ascending order.

On the same data, perform selection sort, but sort data in descending order this time.

Q.No.11:

Write a program to calculate students' average test scores and their grades.

You may assume the following input data:

Johnson 85 83 77 91 76

Aniston 80 90 95 93 48

Cooper 78 81 11 90 73

Gupta 92 83 30 69 87

Blair 23 45 96 38 59

Clark 60 85 45 39 67

Kennedy 77 31 52 74 83

Bronson 93 94 89 77 97

Sunny 79 85 28 93 82

Smith 85 72 49 75 63

Use three arrays: a one-dimensional array to store the students' names, a (parallel) two-dimensional array to store the test scores, and a parallel one-dimensional array to store grades. Your program must contain at least the following functions: a function to read and store data into two arrays, a function to calculate the average test score and grade, and a function to output the results.

Have your program also output the class average.

Q.No.12:

A 4 Byte integer can store at most a 10-digit number. However, in some applications we may want to process huge integers. We can represent such huge integers by keeping each digit of the huge integer as one element of an integer array. For example, the number 23568978965412323698, can be kept as an integer array of 20 elements.

Write a C++ program to compare two huge integers. Your program should work as follows: input two huge integers A and B from the user as character arrays, convert each digit character to its equivalent integer value and store it as an integer array element.

Finally, output 1 if $A > B$, 0 if $A=B$ and -1 if $A < B$. Both the huge integers may or may not have equal number of digits.

However, you can assume that the input huge integer can be of at most 50 digits.

Q.No.13:

An organization wants a system which can keep records of the employee's identification number (or ID) and their salaries. You need to create a prototype for the organization which can store records of 50 employees and allows the user to perform the following operations.

a) Insert a record (take an employee's 4-digit integer ID & his/her salary - a float value)

- Ensure that a user enters a unique 4-digit employee ID, and the salary must be greater than Rs. 20,000.

b) Search a record by ID (search and display an employee's record by ID and display his/her salary or display a message if there is no matching record)

c) Search records by Salary (search and display employees' records by salary and display IDs of all the employees having that salary or display a message if there is no matching record)

d) Delete a record (delete the record of a specific employee by ID or display a message if no record exists)

- Do not forget to delete salary record of that employee in the other array at the corresponding index

e) Display all records (print IDs and salaries of all the employees or display a message if there is no record)

Hints:

- You need two separate arrays (one for employees' ID and another for salaries).

Corresponding indexes of both arrays will represent an individual employee's record. That is, index Zero of both arrays will store ID and salary of first employee, index One of both arrays will store ID and salary of second employee, and so on. This concept is known as 'Parallel arrays'.

- Initially assign -1 to both arrays to represent empty cells. It will also help you while deleting a record.
- You may want to maintain a count variable to keep track of the number of records entered by the user so far. It will help you to decide where to insert a new record.
- Note that a user may delete records from any index. This will leave many empty cells in between.
- If we do not utilize those empty cells for new record insertion, it will be a huge wastage of memory. Think carefully that how you can utilize those empty cells

Q.No.14:

Write a program that can be used to assign seats for a commercial airplane. The airplane has 13 rows with six seats in each row. Rows 1 and 2 are first class, rows 3 through 7 are business class, and rows 8 through 13 are economy class. Followings are the prices of each seat type whereas the price of a seat increases by 5% if that is a window seat (that is, seats in columns A and F).

Class	Price
Economy Class	Rs. 40,000
Business Class	Rs. 85,000
First Class	Rs. 150,000

Write a menu-driven C++ program which allows the following operations.

1. Seat reservation
2. View seating plan
3. Quit

Your program should prompt the user to enter the following information if he/she wants to reserve seat(s).

- a) Ticket type (first class, business class, or economy class – use the last letter for each seat type as input, i.e., 'y' for economy class, 's' for business class and 't' for first class)
- b) Number of seats (must be greater than zero and less than 10)

c) Desired seat (allow the user to select a seat by entering row number (an integer value from 1 to 13) and column name (a character from A to F), as shown below in the figure) Your program should keep taking inputs from the user if any of the input is incorrect, e.g., if a user selects a wrong seat or an already occupied seat, take the input again. Compute and display the total price to the user upon successful seat reservation and direct the user to the main menu.

We assume that a user can only reserve seats of a specific class in one transaction.

However, note that a user may not necessarily selects consecutive seats. For example, if someone wants to buy three economy class seats, he/she may select the following available seats: 5E, 5F and 6E which costs Rs. 122,000 in total.

- ‘View seating plan’ option will display the seating plan as shown in the following image. Here, * indicates that the seat is ‘available’; X indicates that the seat is ‘occupied’. Show the columns and rows names as well. Use tabs or setw manipulator to format your output in a nice way.

- Your program should only terminate if a user selects the ‘Quit’ option from the main menu.

Before terminating, your program should save the seating plan (in tabular form) in a text file and loads the recently saved seating plan from the same file in the subsequent program execution(s).

	A	B	C	D	E	F
Row 1	*	*	X	*	X	X
Row 2	*	X	*	X	*	X
Row 3	*	*	X	X	*	X
Row 4	X	*	X	*	X	X
Row 5	*	X	*	X	*	*
Row 6	*	X	*	*	*	X
Row 7	X	*	*	*	X	X
Row 8	*	X	*	X	X	*
Row 9	X	*	X	X	*	X
Row 10	*	X	*	X	X	X
Row 11	*	*	X	*	X	*
Row 12	*	*	X	X	*	X
Row 13	*	*	*	*	X	*

Q.No.15:

Take the name of a file as input in form of a string from user, count the total number of words

(using string) in the file and print the count.

For example,

Input: word_file.txt

Output: 103 words

Word_file.txt data:

“It is a long-established fact that a reader will be distracted by the readable content of a page when looking at its layout. The point of using Lorem Ipsum is that it has a more-or-less normal distribution of letters, as opposed to using 'Content here, content here', making it look like

readable English. Many desktop publishing packages and web page editors now use Lorem Ipsum as their default model text, and a search for 'lorem ipsum' will uncover many web sites still in their infancy. Various versions have evolved over the years, sometimes by accident, sometimes on purpose (injected humor and the like).”

Q.No.16:

Write a C++ Program to Merge Two Files into a Single file using File Handling. First File should contain half of your intro and the second file should contain the remaining half of your intro. The resulting Third file will contain your full intro. Full intro Should contain, your name, father name, city, roll number, email, hobbies, favorite movies, favorite Subject, etc.

Note: Make the program generic, it should work either I store whole into one line or multiple lines.

Example Input Files:

Bio1.txt	Bio2.txt
Name XYZ	Father Name XYZ
City XYZ	Roll Number XYZ
Email XYZ	Hobbies
Favorite Movies XYZ	Favorite Subject

Example Output File:

Name XYZ

Father Name
XYZ City XYZ
Roll Number
XYZ Email
XYZ Hobbies
Favorite Movies
XYZ Favorite
Subject

Q.No.17:

Given an unsorted array **Arr** of size **N** of positive integers. **One number 'A'** from set {1, 2,...,N} is missing and **one number 'B'** occurs twice in array. Find these two number

Q.No.18:

Write a c++ code, make a function in which user input 23 function should return 24 and when user insert 24 function should return 23 **i.e.** do not use if else

Q.No.19:

Given two arrays **a[]** and **b[]** of size **n** and **m** respectively. The task is to find the number of elements in the union between these two arrays.

Union of the two arrays can be defined as the set containing distinct elements from both the arrays. If there are repetitions, then only one occurrence of element should be printed in the union.

Note : Elements are not necessarily distinct.

Q.No.20:

Given a string **S**, check if it is palindrome or not.

Input: S = "abba"

Output: 1

Explanation: S is a palindrome

Input: S = "abc"

Output: 0

Explanation: S is not a palindrome

Q.No.21:

Given a string S, find the longest palindromic substring in S. **Substring of string S:** $S[i \dots j]$ where $0 \leq i \leq j < \text{len}(S)$. **Palindrome string:** A string that reads the same backward. More formally, S is a palindrome if $\text{reverse}(S) = S$. **In case of conflict**, return the substring which occurs first (with the least starting index).

Input:

S = "aaaabbbaa"

Output: aabbbaa

Explanation: The longest Palindromic substring is "aabbbaa"

Q.No.22:

Given a **sorted** array of **positive** integers. Your task is to rearrange the array elements alternatively i.e first element should be max value, second should be min value, third should be second max, fourth should be second min and so on.

Note: Modify the original array itself. Do it without using any extra space. You do not have to return anything.

Example:

Input:

n = 6

arr[] = {1,2,3,4,5,6}

Output: 6 1 5 2 4 3

Explanation: Max element = 6, min = 1, second max = 5, second min = 2, and so on... Modified array is : 6 1 5 2 4 3.

Q.No.23:

Given an array arr[] of size n, find the first repeating element. The element should occur more than once and the index of its first occurrence should be the smallest.

Note:- The position you return should be according to 1-based indexing.

Example

Input:

n = 7

arr[] = {1, 5, 3, 4, 3, 5, 6}

Output: 2

Explanation:

5 is appearing twice and its first appearance is at index 2 which is less than 3 whose first occurring index is 3.

Q.No.24:

Given two numbers as strings s1 and s2. Calculate their Product.

Note: The numbers can be negative and You are not allowed to use any built-in function or convert the strings to integers. There can be zeros in the beginning of the numbers. You don't need to specify '+' sign in the beginning of positive numbers.

Example 1:

Input:

s1 = "0033"

s2 = "2"

Output:

66

Q.No.25:

Given an array **arr[]** of size **N**, check if it is sorted in non-decreasing order or not.

Example 1:

Input:

N = 5

arr[] = {10, 20, 30, 40, 50}

Output: 1

Explanation: The given array is sorted

Q.No.26:

You are given an array **A** of size **N**. You need to print elements of A in alternate order (starting from index 0).

Example 2:

Input:

$N = 5$

$A[] = \{1, 2, 3, 4, 5\}$

Output:

1 3 5

Q.No.27:

You are given an integer N. You need to convert all zeroes of N to 5.

Example 1:

Input:

N = 1004

Output: 1554

Q.No.28:

Given a list of names, display the longest name.

Example:

Input:

N = 5

names[] = { "Geek", "Geeks", "Geeksfor",
"GeeksforGeek", "GeeksforGeeks" }

Output:

GeeksforGeeks

Q.No.29:

Given a string, remove spaces from it.

Example 1:

Input:

S = "geeks for geeks"

Output: geeksforgeeks

Explanation: All the spaces have been removed.

Q.No.30:

Given two arrays **A** and **B** contains integers of size **N** and **M**, the task is to find numbers which are present in the first array, but not present in the second array.

Example 1:**Input:** N = 6, M = 5

A[] = {1, 2, 3, 4, 5, 10}

B[] = {2, 3, 1, 0, 5}

Output: 4 10**Q.No.31:**

Given an array of size **N** and you have to tell whether the array is perfect or not. An array is said to be perfect if its reverse array matches the original array. If the array is perfect then **return True else return False.**

Input : Arr[] = {1, 2, 3, 2, 1}**Output :** PERFECT**Q.No.32:**

Given an unsigned integer **N**. The task is to swap all odd bits with even bits. For example, if the given number is 23 (**00010111**), it should be converted to 43(**00101011**). Here, every even position bit is swapped with an adjacent bit on the right side(even position bits are highlighted in the binary representation of 23), and every odd position bit is swapped with an adjacent on the left side.

Example 1:**Input:** N = 23**Output:** 43**Explanation:**

Binary representation of the given number is **00010111** after swapping
00101011 = 43 in decimal.

Q.No.33:

Given a number num, our task is to find the closest Palindrome number whose absolute difference with given number is minimum. If 2 Palindrome numbers have same absolute difference from the given number, then find the smaller one.

Example 2:**Input:** num = 489

Output: 484

Q.No.34:

Given an even number **N** (greater than 2), return two prime numbers whose sum will be equal to given number. There are several combinations possible. Print only the pair whose minimum value is the smallest among all the minimum values of pairs and print the minimum element first.

Example 1:

Input: N = 74

Output: 3 71

Explanation: There are several possibilities like 37 37. But the minimum value of this pair is 3 which is smallest among all possible minimum values of all the pairs.

Q.No.35:

Given a string **S** which consists of only lowercase English alphabets, you have to perform the below operations:

If the string **S** contains any repeating character, remove the first repeating character and reverse the string and again perform the above operation on the modified string, otherwise, you stop.

You have to find the final string.

Example 1:

Input: S = "abab"

Output: ba

Q.No.36:

Given a string **S** of **lowercase** english characters. Rearrange characters of the given string such that the vowels and consonants occupy **alternate** positions and the string so formed should be **lexicographically** (alphabetically) **smallest**.

Note: Vowels are 'a', 'e', 'i', 'o' and 'u'.

Example 1:

Input:

S = "aeroplane"

Output: alanepero

Q.No.37:

Given a number, reverse it and add it to itself unless it becomes a palindrome or number of iterations becomes more than 5.

Q.No.38:

Given a string **str** containing only lowercase letters, generate a string with the same letters, but in uppercase.

Example 1:

Input:

str = "geeks"

Output: GEEKS

Q.No.39:

Count the numbers between 1 to **N** containing 4 as a digit

Example 2:

Input:

N = 14

Output:

2

Q.No.40:

You are given four numbers **num1**, **den1**, **num2**, and **den2**. You need to find **(num1/den1)+(num2/den2)** and output the result in the form of **(numx/denx)**.

Q.No.41:

Given an unsorted array **arr** of size **N**, rearrange the elements of array such that number at the **odd** index is greater than the number at the **previous even** index. The task is to complete the function **formatArray()** which will return formatted array.

Example 1:

Input:

n = 5

arr[] = {5, 4, 3, 2, 1}

Q.No.42:

Print this pattern

Input: 5

Output:

ABCDE

ABCD

ABC

AB

A

Q.No.43:

Given a string **str** containing only lower case alphabets, the task is to sort it in lexicographically-descending order.

Input: str = "for"

Output: "rof"

Q.No.44:

A prime number is Super Prime if it is a sum of two primes. Find all the Super Primes upto N

Input:

N = 5

Output: 1

Explanation: $5 = 2 + 3$, 5 is the only super prime

Q.No.45:

Print this pattern

Input: 5

Output:

A

BB

CCC

DDDD

EEEE

Q.No.46:

Given a string containing uppercase alphabets and integer digits (from 0 to 9), the task is to print the alphabets in the lexicographical order followed by the sum of digits

Input: S = "AC2BEW3"

Output: "ABCEW5"

Explanation: $2 + 3 = 5$ and we print all

Q.No.47:

Given an integer N represented in the form of a string, remove consecutive repeated digits from it.

Input:

N = 1224

Output: 124

Q.No.48:

Given a number positive number N, find value of $f_0 + f_1 + f_2 + \dots + f_N$ where f_i indicates ith Fibonacci number.

Remember that $f_0 = 0$, $f_1 = 1$, $f_2 = 1$, $f_3 = 2$, $f_4 = 3$, $f_5 = 5$,

Input:

N = 3

Output:

4

Explanation:

$0 + 1 + 1 + 2 = 4$

Q.No.49:

Given a non-negative integer N, repeatedly add all its digits until the result has only one digit.

Input:

N = 98

Output:

8

Explanation:

Sum of digit of N = 98 is $9 + 8 = 17$

Sum of digit of 17 = $1 + 7 = 8$

Since 8 is the single digit number hence it is the required answer.

Q.No.50:

Find the count of numbers less than equal to N having exactly 9 divisors.

Input:

N = 100

Output:

2

Explanation:

The two numbers which have exactly 9 divisors are 36 and 100.